

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Assisted senior engineers with **site surveys** and **data collection** for industrial client energy assessments, contributing to baseline efficiency reports.
- Analyzed **energy consumption data** using **Python** and advanced **Excel**, identifying key inefficiencies and proposing 15+ actionable energy-saving opportunities.
- Supported the preparation of technical reports, including **engineering calculations** and preliminary design documentation for HVAC and steam systems.
- Gained hands-on exposure to core industrial utility systems, specifically documenting operational parameters for **HVAC**, **Steam**, and **Compressed Air** systems.
- Contributed to energy audits and baseline development, aligning data collection methodologies with standards like **IS 399** and **ISO 50001**.
- Participated in client meetings and site visits, developing confidence in technical communication and stakeholder engagement across multiple industrial sectors.

SKILLS

Engineering Software – MATLAB, AutoCAD, Revit, Energy Modelling Tools

Data Analysis & Tools – Python, Pandas, NumPy, Excel (Advanced), R

Energy Standards & Regulations – IS 399, EN 16247, ISO 50001, IPMVP, SEAI Energy Programmes

Industrial Systems – HVAC, Steam, Chilled Water, LPHW, Compressed Air, Cooling Towers, Electrical Distribution

Core Engineering Skills – Energy Assessments, Site Surveys, Data Collection, Technical Reporting, Feasibility Studies, Engineering Calculations, Design Documentation, Measurement Verification, Project Scoping, Cost Estimation

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Industrial Energy Efficiency Analysis Dashboard

- Developed a **Power BI** dashboard to visualize energy consumption patterns across client facilities, integrating data from multiple sources.
- Designed and implemented **DAX measures** for key performance indicators (KPIs) related to energy usage intensity and cost savings.

- Automated data extraction and transformation processes using **Power Query**, reducing manual data preparation time by 40% for reporting cycles.

Energy Systems Monitoring and Reporting Tool

- Built a **Python** application to ingest and process real-time operational data from simulated industrial utility systems.
- Implemented REST API endpoints using **Flask** for data retrieval, enabling integration with a separate reporting module.
- Configured **SQLAlchemy** for database interactions, storing time-series data for historical analysis and trend identification.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)